

ROADRESQ

Database Schema Document

Version 1.0

PostgreSQL + PostGIS • Relational Data Model

REQUEST → MATCH → TRACK → REPAIR → PAY → REVIEW

Production-oriented database blueprint for the RoadResQ roadside assistance and automotive service platform.

Document	RoadResQ Database Schema
Version	1.0
Database	PostgreSQL
Geospatial	PostGIS
Primary scope	Customer, Provider, Booking, Job, Estimate, Invoice, Payment, Review

1. Database Design Objectives

RoadResQ uses a normalized relational model to support emergency roadside assistance, scheduled vehicle service, provider discovery, live job progression, inspection, estimates, invoices, payments, reviews, support, and administration.

Objective	Design decision
Data integrity	Primary/foreign keys, NOT NULL, CHECK, UNIQUE, and transactional updates.
Location search	PostGIS geography fields with GiST spatial indexes.
Traceability	Status history, audit logs, payment records, and provider location history.
Scalability	UUIDs, indexed foreign keys, modular domain boundaries, and retention policies.
Security	Access-controlled documents and financial references; raw card data is never stored.

2. High-Level Entity Model

User → Vehicle → Booking → Provider → Job Card → Estimate → Invoice → Payment → Review

Domain	Core entities
Identity	users, user_addresses
Vehicles	vehicles
Provider	providers, provider_documents, provider_services, provider_availability, provider_location_updates
Catalog	services
Bookings	bookings, booking_locations, booking_status_history
Job execution	job_cards, inspections, inspection_items, estimates, estimate_items
Billing	invoices, invoice_items, payments, payouts
Engagement	reviews, notifications, messages, support_tickets
Governance	audit_logs

3. Common Data Conventions

Convention	Standard
ID	UUID primary keys.
Timestamps	TIMESTAMPPTZ in UTC.
Money	NUMERIC(12,2) plus currency code.
Status	Controlled values validated at application and database layers.
Location	GEOGRAPHY(Point,4326).
Audit	created_at and updated_at; actor IDs where needed.

4. Identity & Account Schema

4.1 users

Purpose: All platform accounts.

Column	Type	Key / Null	Description
id	UUID	PK	Unique user identifier.
role	VARCHAR(30)	NOT NULL	CUSTOMER, PROVIDER, ADMIN.
first_name	VARCHAR(80)	NOT NULL	Given name.
last_name	VARCHAR(80)	NULL	Family name.
phone	VARCHAR(20)	UNIQUE	Primary contact identity.
email	VARCHAR(255)	UNIQUE	Optional email.
password_hash	TEXT	NULL	Password hash if enabled.
status	VARCHAR(30)	NOT NULL	ACTIVE, SUSPENDED, BLOCKED, PENDING.
last_login_at	TIMESTAMPTZ	NULL	Last successful login.
created_at	TIMESTAMPTZ	NOT NULL	Creation.
updated_at	TIMESTAMPTZ	NOT NULL	Last update.

Indexes: UNIQUE(phone), UNIQUE(email), INDEX(role,status)

Rules: Never store plaintext OTPs or passwords.

4.2 user_addresses

Purpose: Saved customer addresses.

Column	Type	Key / Null	Description
id	UUID	PK	Address identifier.
user_id	UUID	FK → users	Owner.
label	VARCHAR(40)	NOT NULL	Home, Work, Other.
address_line1	TEXT	NOT NULL	Street/building.
address_line2	TEXT	NULL	Additional details.
city	VARCHAR(100)	NOT NULL	City.
state	VARCHAR(100)	NULL	State/region.
postal_code	VARCHAR(20)	NULL	Postal code.
location	GEOGRAPHY(Point,4326)	NULL	Geocoded point.
is_default	BOOLEAN	NOT NULL	Default flag.

Indexes: INDEX(user_id), GIST(location)

Rules: Allow only one default address per user.

5. Vehicle Schema

5.1 vehicles

Purpose: Vehicles owned or managed by customers.

Column	Type	Key / Null	Description
id	UUID	PK	Vehicle identifier.
user_id	UUID	FK → users	Vehicle owner.
registration_number	VARCHAR(30)	UNIQUE	Registration number.
make	VARCHAR(80)	NULL	Manufacturer.
model	VARCHAR(100)	NULL	Model.
variant	VARCHAR(100)	NULL	Variant/trim.
fuel_type	VARCHAR(30)	NULL	PETROL, DIESEL, CNG, EV, HYBRID, OTHER.

Column	Type	Key / Null	Description
year	SMALLINT	NULL	Model/manufacturing year.
color	VARCHAR(50)	NULL	Color.
vin	VARCHAR(100)	NULL	VIN/chassis identifier.
odometer_km	INTEGER	NULL	Latest odometer.
is_primary	BOOLEAN	NOT NULL	Primary vehicle.
created_at	TIMESTAMPTZ	NOT NULL	Creation.
updated_at	TIMESTAMPTZ	NOT NULL	Last update.

Indexes: UNIQUE(registration_number), INDEX(user_id)

Rules: Normalize registration numbers before uniqueness checks.

6. Service & Provider Schema

6.1 services

Purpose: Master service catalog.

Column	Type	Key / Null	Description
id	UUID	PK	Service identifier.
name	VARCHAR(120)	UNIQUE	Service name.
category	VARCHAR(50)	NOT NULL	TOWING, BATTERY, TYRE, FUEL, MECHANICAL, GENERAL, SERVICE.
description	TEXT	NULL	Customer-facing description.
base_price	NUMERIC(12,2)	NULL	Reference price.
is_emergency	BOOLEAN	NOT NULL	Allowed in emergency flow.
is_active	BOOLEAN	NOT NULL	Catalog active.

Indexes: INDEX(category,is_active)

6.2 providers

Purpose: Garages, mechanics, towing operators, and service providers.

Column	Type	Key / Null	Description
id	UUID	PK	Provider identifier.
user_id	UUID	FK → users	Account owner.
business_name	VARCHAR(160)	NOT NULL	Public business name.
provider_type	VARCHAR(40)	NOT NULL	MECHANIC, GARAGE, TOWING, BATTERY, TYRE, OTHER.
description	TEXT	NULL	Business description.
phone	VARCHAR(20)	NULL	Business contact.
service_radius_km	NUMERIC(6,2)	NOT NULL	Dispatch radius.
rating_avg	NUMERIC(3,2)	NOT NULL	Cached rating.
rating_count	INTEGER	NOT NULL	Cached review count.
verification_status	VARCHAR(30)	NOT NULL	PENDING, VERIFIED, ACTIVE, SUSPENDED, REJECTED.
is_online	BOOLEAN	NOT NULL	Current availability.
created_at	TIMESTAMPTZ	NOT NULL	Creation.
updated_at	TIMESTAMPTZ	NOT NULL	Last update.

Indexes: UNIQUE(user_id), INDEX(verification_status,is_online), INDEX(provider_type)

Rules: Dispatch only VERIFIED/ACTIVE providers.

6.3 provider_services

Purpose: Maps providers to supported services.

Column	Type	Key / Null	Description
id	UUID	PK	Mapping identifier.
provider_id	UUID	FK → providers	Provider.
service_id	UUID	FK → services	Service.
price_from	NUMERIC(12,2)	NULL	Starting/reference price.
price_to	NUMERIC(12,2)	NULL	Upper/reference price.
is_active	BOOLEAN	NOT NULL	Currently offered.

Indexes: UNIQUE(provider_id,service_id), INDEX(service_id,is_active)

6.4 provider_availability

Purpose: Recurring provider working windows.

Column	Type	Key / Null	Description
id	UUID	PK	Availability identifier.
provider_id	UUID	FK → providers	Provider.
day_of_week	SMALLINT	NOT NULL	Consistent 0–6 or 1–7 convention.
start_time	TIME	NOT NULL	Start.
end_time	TIME	NOT NULL	End.
is_active	BOOLEAN	NOT NULL	Active rule.

Indexes: INDEX(provider_id,day_of_week)

7. Provider Verification & Location

7.1 provider_documents

Purpose: Verification metadata and secure file references.

Column	Type	Key / Null	Description
id	UUID	PK	Document identifier.
provider_id	UUID	FK → providers	Provider.
document_type	VARCHAR(50)	NOT NULL	IDENTITY, ADDRESS, BUSINESS, LICENSE, OTHER.
file_url	TEXT	NOT NULL	Private object-storage reference.
document_number	VARCHAR(100)	NULL	Optional number.
status	VARCHAR(30)	NOT NULL	PENDING, APPROVED, REJECTED, EXPIRED.
reviewed_by	UUID	FK → users	Admin reviewer.
reviewed_at	TIMESTAMPTZ	NULL	Review time.
created_at	TIMESTAMPTZ	NOT NULL	Upload time.

Indexes: INDEX(provider_id,status), INDEX(document_type)

Rules: Store files outside PostgreSQL; database keeps metadata/reference.

7.2 provider_location_updates

Purpose: Historical GPS points for active jobs.

Column	Type	Key / Null	Description
id	UUID	PK	Location event.
provider_id	UUID	FK → providers	Provider.
booking_id	UUID	FK → bookings	Active job.
location	GEOGRAPHY(Point,4326)	NOT NULL	GPS point.
accuracy_m	NUMERIC(8,2)	NULL	Reported accuracy.
heading	NUMERIC(6,2)	NULL	Heading.
speed_kmh	NUMERIC(7,2)	NULL	Approximate speed.
recorded_at	TIMESTAMPTZ	NOT NULL	Device timestamp.

Indexes: GIST(location), INDEX(booking_id,recorded_at), INDEX(provider_id,recorded_at)

Rules: Apply retention rules to historical GPS points.

8. Booking & Dispatch Schema

8.1 bookings

Purpose: Central transaction for emergency and scheduled service.

Column	Type	Key / Null	Description
id	UUID	PK	Booking identifier.
booking_number	VARCHAR(30)	UNIQUE	Human-readable reference.
customer_id	UUID	FK → users	Customer.
vehicle_id	UUID	FK → vehicles	Vehicle.
provider_id	UUID	FK → providers	Assigned provider; null while searching.
service_id	UUID	FK → services	Requested service.
booking_type	VARCHAR(30)	NOT NULL	EMERGENCY or SCHEDULED.
status	VARCHAR(40)	NOT NULL	Current lifecycle state.
problem_description	TEXT	NULL	Customer description.
scheduled_at	TIMESTAMPTZ	NULL	Scheduled time.

Column	Type	Key / Null	Description
requested_at	TIMESTAMPTZ	NOT NULL	Request time.
accepted_at	TIMESTAMPTZ	NULL	Acceptance.
arrived_at	TIMESTAMPTZ	NULL	Arrival.
completed_at	TIMESTAMPTZ	NULL	Completion.
cancelled_at	TIMESTAMPTZ	NULL	Cancellation.
cancellation_reason	TEXT	NULL	Cancellation reason.
created_at	TIMESTAMPTZ	NOT NULL	Creation.
updated_at	TIMESTAMPTZ	NOT NULL	Last update.

Indexes: UNIQUE(booking_number), INDEX(customer_id,created_at), INDEX(provider_id,status), INDEX(status,requested_at), INDEX(scheduled_at)

Rules: Provider assignment must be transactional to prevent double-claiming.

8.2 booking_locations

Purpose: Pickup/service/destination points.

Column	Type	Key / Null	Description
id	UUID	PK	Location identifier.
booking_id	UUID	FK → bookings	Booking.
location_type	VARCHAR(30)	NOT NULL	PICKUP, SERVICE, DESTINATION.
address_text	TEXT	NULL	Display address.
location	GEOGRAPHY(Point,4326)	NOT NULL	Coordinates.
landmark	TEXT	NULL	Optional landmark.
created_at	TIMESTAMPTZ	NOT NULL	Creation.

Indexes: GIST(location), INDEX(booking_id,location_type)

8.3 booking_status_history

Purpose: Immutable booking state transitions.

Column	Type	Key / Null	Description
id	UUID	PK	History record.
booking_id	UUID	FK → bookings	Booking.
from_status	VARCHAR(40)	NULL	Previous state.
to_status	VARCHAR(40)	NOT NULL	New state.
changed_by	UUID	FK → users	Actor.
note	TEXT	NULL	Transition note.
created_at	TIMESTAMPTZ	NOT NULL	Transition time.

Indexes: INDEX(booking_id,created_at)

Rules: Append-only history; do not overwrite old transitions.

9. Job Execution Schema

9.1 job_cards

Purpose: Operational work order after provider acceptance/arrival.

Column	Type	Key / Null	Description
id	UUID	PK	Job card identifier.
booking_id	UUID	FK → bookings	Related booking.
provider_id	UUID	FK → providers	Provider.
mechanic_user_id	UUID	FK → users	Specific mechanic, optional.
job_status	VARCHAR(40)	NOT NULL	INSPECTION, ESTIMATE_PENDING, APPROVED, IN_PROGRESS, COMPLETED.
started_at	TIMESTAMPZ	NULL	Work start.
completed_at	TIMESTAMPZ	NULL	Completion.
customer_notes	TEXT	NULL	Customer-facing notes.
provider_notes	TEXT	NULL	Provider notes.
created_at	TIMESTAMPZ	NOT NULL	Creation.
updated_at	TIMESTAMPZ	NOT NULL	Last update.

Indexes: UNIQUE(booking_id), INDEX(provider_id,job_status)

9.2 inspections

Purpose: Vehicle inspection session.

Column	Type	Key / Null	Description
id	UUID	PK	Inspection identifier.
job_card_id	UUID	FK → job_cards	Job.
odometer_km	INTEGER	NULL	Inspection odometer.
overall_notes	TEXT	NULL	Summary.
created_by	UUID	FK → users	Inspector.
created_at	TIMESTAMPZ	NOT NULL	Inspection time.

Indexes: INDEX(job_card_id)

9.3 inspection_items

Purpose: Individual inspection findings.

Column	Type	Key / Null	Description
id	UUID	PK	Inspection item.
inspection_id	UUID	FK → inspections	Inspection.
component	VARCHAR(100)	NOT NULL	Component inspected.
condition	VARCHAR(40)	NOT NULL	GOOD, FAIR, POOR, CRITICAL.
finding	TEXT	NULL	Finding.
media_url	TEXT	NULL	Photo/video reference.
recommended_action	TEXT	NULL	Recommended work.

Indexes: INDEX(inspection_id)

10. Estimates & Billing Schema

10.1 estimates

Purpose: Customer approval document for proposed work.

Column	Type	Key / Null	Description
id	UUID	PK	Estimate identifier.

Column	Type	Key / Null	Description
job_card_id	UUID	FK → job_cards	Job.
estimate_number	VARCHAR(30)	UNIQUE	Reference.
subtotal	NUMERIC(12,2)	NOT NULL	Before tax/discount.
tax_amount	NUMERIC(12,2)	NOT NULL	Tax.
discount_amount	NUMERIC(12,2)	NOT NULL	Discount.
total_amount	NUMERIC(12,2)	NOT NULL	Final estimate.
status	VARCHAR(30)	NOT NULL	DRAFT, SENT, APPROVED, REJECTED, EXPIRED.
approved_by	UUID	FK → users	Customer approver.
approved_at	TIMESTAMPTZ	NULL	Approval time.
created_at	TIMESTAMPTZ	NOT NULL	Creation.

Indexes: INDEX(job_card_id,status)

10.2 estimate_items

Purpose: Estimate line items.

Column	Type	Key / Null	Description
id	UUID	PK	Line item.
estimate_id	UUID	FK → estimates	Estimate.
description	TEXT	NOT NULL	Work/part description.
quantity	NUMERIC(10,2)	NOT NULL	Quantity.
unit_price	NUMERIC(12,2)	NOT NULL	Unit price.
tax_rate	NUMERIC(5,2)	NOT NULL	Tax percentage.
line_total	NUMERIC(12,2)	NOT NULL	Calculated line total.

Indexes: INDEX(estimate_id)

Rules: Store a snapshot of the price/tax values used.

10.3 invoices

Purpose: Final financial document.

Column	Type	Key / Null	Description
id	UUID	PK	Invoice identifier.
booking_id	UUID	FK → bookings	Booking.
invoice_number	VARCHAR(40)	UNIQUE	Invoice reference.
subtotal	NUMERIC(12,2)	NOT NULL	Subtotal.
tax_amount	NUMERIC(12,2)	NOT NULL	Tax.
discount_amount	NUMERIC(12,2)	NOT NULL	Discount.
total_amount	NUMERIC(12,2)	NOT NULL	Amount due.
currency	CHAR(3)	NOT NULL	e.g. INR.
status	VARCHAR(30)	NOT NULL	DRAFT, ISSUED, PARTIALLY_PAID, PAID, VOID.
issued_at	TIMESTAMPTZ	NULL	Issue time.
created_at	TIMESTAMPTZ	NOT NULL	Creation.

Indexes: UNIQUE(booking_id), UNIQUE(invoice_number), INDEX(status,issued_at)

10.4 invoice_items

Purpose: Final invoice line-item snapshot.

Column	Type	Key / Null	Description
id	UUID	PK	Line item.
invoice_id	UUID	FK → invoices	Invoice.

Column	Type	Key / Null	Description
description	TEXT	NOT NULL	Work/part/service.
quantity	NUMERIC(10,2)	NOT NULL	Quantity.
unit_price	NUMERIC(12,2)	NOT NULL	Unit price.
tax_rate	NUMERIC(5,2)	NOT NULL	Tax rate.
line_total	NUMERIC(12,2)	NOT NULL	Line total.

Indexes: INDEX(invoice_id)

11. Payments & Payouts

11.1 payments

Purpose: Customer payment attempts and successful transactions.

Column	Type	Key / Null	Description
id	UUID	PK	Payment identifier.
invoice_id	UUID	FK → invoices	Invoice.
customer_id	UUID	FK → users	Payer.
amount	NUMERIC(12,2)	NOT NULL	Payment amount.
currency	CHAR(3)	NOT NULL	Currency.
method	VARCHAR(30)	NOT NULL	UPI, CASH, CARD, NET_BANKING, OTHER.
status	VARCHAR(30)	NOT NULL	PENDING, SUCCESS, FAILED, REFUNDED.
gateway	VARCHAR(50)	NULL	Gateway/provider.
gateway_transaction_id	VARCHAR(150)	NULL	External reference.
paid_at	TIMESTAMPZ	NULL	Successful payment time.
created_at	TIMESTAMPZ	NOT NULL	Creation.

Indexes: INDEX(invoice_id,status), INDEX(gateway_transaction_id), INDEX(customer_id,created_at)

Rules: Never store card number, CVV, or raw payment credentials.

11.2 payouts

Purpose: Provider settlement records.

Column	Type	Key / Null	Description
id	UUID	PK	Payout identifier.
provider_id	UUID	FK → providers	Payee.
booking_id	UUID	FK → bookings	Source booking.
amount	NUMERIC(12,2)	NOT NULL	Provider payout.
platform_fee	NUMERIC(12,2)	NOT NULL	Platform fee.
status	VARCHAR(30)	NOT NULL	PENDING, PROCESSING, PAID, FAILED.
external_reference	VARCHAR(150)	NULL	Settlement reference.
paid_at	TIMESTAMPZ	NULL	Paid time.
created_at	TIMESTAMPZ	NOT NULL	Creation.

Indexes: INDEX(provider_id,status), INDEX(booking_id)

12. Reviews, Notifications & Communication

12.1 reviews

Purpose: Customer feedback for completed provider jobs.

Column	Type	Key / Null	Description
id	UUID	PK	Review identifier.
booking_id	UUID	FK → bookings	Completed booking.
customer_id	UUID	FK → users	Reviewer.
provider_id	UUID	FK → providers	Reviewed provider.
rating	SMALLINT	NOT NULL	1–5.
comment	TEXT	NULL	Optional text.
created_at	TIMESTAMPZ	NOT NULL	Review time.

Indexes: UNIQUE(booking_id,customer_id), INDEX(provider_id,created_at)

12.2 notifications

Purpose: In-app notification events.

Column	Type	Key / Null	Description
id	UUID	PK	Notification identifier.
user_id	UUID	FK → users	Recipient.
booking_id	UUID	FK → bookings	Optional booking.
type	VARCHAR(50)	NOT NULL	Event type.
title	VARCHAR(160)	NOT NULL	Title.
body	TEXT	NOT NULL	Body.
is_read	BOOLEAN	NOT NULL	Read state.
created_at	TIMESTAMPTZ	NOT NULL	Creation.
read_at	TIMESTAMPTZ	NULL	Read time.

Indexes: INDEX(user_id,is_read,created_at)

12.3 messages

Purpose: Customer-provider job communication.

Column	Type	Key / Null	Description
id	UUID	PK	Message identifier.
booking_id	UUID	FK → bookings	Related job.
sender_id	UUID	FK → users	Sender.
recipient_id	UUID	FK → users	Recipient.
message_type	VARCHAR(30)	NOT NULL	TEXT, IMAGE, SYSTEM.
content	TEXT	NULL	Message body.
media_url	TEXT	NULL	Secure media reference.
created_at	TIMESTAMPTZ	NOT NULL	Sent time.

Indexes: INDEX(booking_id,created_at), INDEX(recipient_id,created_at)

13. Support, Audit & Administration

13.1 support_tickets

Purpose: Issues, disputes, and support requests.

Column	Type	Key / Null	Description
id	UUID	PK	Ticket identifier.
ticket_number	VARCHAR(30)	UNIQUE	Reference.
created_by	UUID	FK → users	Requester.
booking_id	UUID	FK → bookings	Optional booking.
category	VARCHAR(50)	NOT NULL	PAYMENT, PROVIDER, SERVICE, SAFETY, ACCOUNT, OTHER.
priority	VARCHAR(20)	NOT NULL	LOW, MEDIUM, HIGH, CRITICAL.
status	VARCHAR(30)	NOT NULL	OPEN, IN_PROGRESS, RESOLVED, CLOSED.
subject	VARCHAR(180)	NOT NULL	Subject.
description	TEXT	NOT NULL	Issue description.
assigned_admin_id	UUID	FK → users	Assigned admin.
created_at	TIMESTAMPTZ	NOT NULL	Creation.
updated_at	TIMESTAMPTZ	NOT NULL	Last update.
resolved_at	TIMESTAMPTZ	NULL	Resolution time.

Indexes: UNIQUE(ticket_number), INDEX(status,priority), INDEX(created_by,created_at), INDEX(booking_id)

13.2 audit_logs

Purpose: Immutable security and administrative audit trail.

Column	Type	Key / Null	Description
id	UUID	PK	Audit identifier.
actor_user_id	UUID	FK → users	Actor.
action	VARCHAR(100)	NOT NULL	Action performed.
entity_type	VARCHAR(80)	NOT NULL	Affected entity.
entity_id	UUID	NULL	Affected ID.
old_data	JSONB	NULL	Previous state snapshot.
new_data	JSONB	NULL	New state snapshot.
ip_address	INET	NULL	Request source.
user_agent	TEXT	NULL	Client metadata.
created_at	TIMESTAMPTZ	NOT NULL	Audit time.

Indexes: INDEX(entity_type,entity_id,created_at), INDEX(actor_user_id,created_at)

Rules: Protect audit records from normal application-user modification.

14. Relationship & Cardinality Map

Parent	Child	Cardinality	Meaning
users	vehicles	1:N	A customer can have multiple vehicles.
users	providers	1:0..1	Provider account owns one provider profile.
providers	provider_services	1:N	Provider offers services.
services	provider_services	1:N	Service can be offered by many providers.
users	bookings	1:N	Customer creates bookings.
vehicles	bookings	1:N	Vehicle can have many bookings.
providers	bookings	1:N	Provider handles many bookings.
bookings	booking_locations	1:N	Pickup/service/destination points.

Parent	Child	Cardinality	Meaning
bookings	booking_status_history	1:N	Every transition recorded.
bookings	job_cards	1:0..1	Booking may create one job card.
job_cards	inspections	1:N	Multiple inspection sessions possible.
inspections	inspection_items	1:N	Inspection contains findings.
job_cards	estimates	1:N	Revised estimates possible.
estimates	estimate_items	1:N	Estimate contains line items.
bookings	invoices	1:0..1	Completed booking produces invoice.
invoices	invoice_items	1:N	Invoice contains line items.
invoices	payments	1:N	Supports retries/partial payments.
bookings	reviews	1:0..1	Eligible booking can have one review.

15. Booking State Machine

State	Typical next states
REQUESTED	SEARCHING, CANCELLED
SEARCHING	PROVIDER_ASSIGNED, EXPIRED, CANCELLED
PROVIDER_ASSIGNED	ACCEPTED, EXPIRED, CANCELLED
ACCEPTED	ON_THE_WAY, CANCELLED
ON_THE_WAY	ARRIVED, CANCELLED
ARRIVED	INSPECTION, CANCELLED
INSPECTION	ESTIMATE_PENDING, IN_PROGRESS
ESTIMATE_PENDING	APPROVED, CANCELLED, DISPUTED
APPROVED	IN_PROGRESS
IN_PROGRESS	COMPLETED, DISPUTED
COMPLETED	INVOICE_GENERATED
INVOICE_GENERATED	PAYMENT_COMPLETED
PAYMENT_COMPLETED	CLOSED

16. Core PostgreSQL DDL Skeleton

Representative implementation skeleton; production migrations should add exact constraints, triggers, permissions, retention, and application-specific extensions.

```
CREATE EXTENSION IF NOT EXISTS postgis; CREATE TABLE users ( id UUID PRIMARY KEY, role VARCHAR(30) NOT NULL, first_name VARCHAR(80) NOT NULL, last_name VARCHAR(80), phone VARCHAR(20) NOT NULL UNIQUE, email VARCHAR(255) UNIQUE, password_hash TEXT, status VARCHAR(30) NOT NULL DEFAULT 'ACTIVE', created_at TIMESTAMPTZ NOT NULL DEFAULT now(), updated_at TIMESTAMPTZ NOT NULL DEFAULT now() ); CREATE TABLE bookings ( id UUID PRIMARY KEY, booking_number VARCHAR(30) NOT NULL UNIQUE, customer_id UUID NOT NULL REFERENCES users(id), vehicle_id UUID NOT NULL REFERENCES vehicles(id), provider_id UUID REFERENCES providers(id), service_id UUID NOT NULL REFERENCES services(id), booking_type VARCHAR(30) NOT NULL, status VARCHAR(40) NOT NULL, requested_at TIMESTAMPTZ NOT NULL DEFAULT now(), created_at TIMESTAMPTZ NOT NULL DEFAULT now(), updated_at TIMESTAMPTZ NOT NULL DEFAULT now() ); CREATE INDEX idx_bookings_status_requested ON bookings(status, requested_at); CREATE INDEX idx_booking_locations_gist ON booking_locations USING GIST(location);
```

17. Indexing Strategy

Data	Important indexes	Reason
users	phone, email, role/status	Authentication and filtering.
vehicles	user_id, registration_number	Ownership and lookup.
providers	verification_status/is_online	Dispatch candidate filtering.
provider_services	service_id/is_active	Service matching.
provider_location_updates	GIST(location), booking_id/recorded_at	Spatial and tracking queries.
bookings	status/requested_at, customer_id/created_at, provider_id/status	Dispatch and dashboards.
booking_locations	GIST(location)	Nearby operations.
booking_status_history	booking_id/created_at	Timeline reconstruction.
payments	invoice_id/status, gateway_transaction_id	Reconciliation.
audit_logs	entity_type/entity_id/created_at	Security tracing.

18. Geospatial Design

Use PostGIS **GEOGRAPHY(Point,4326)** for location-aware operations. The primary flow is spatial filtering first, then ranking candidates by distance/ETA and other matching factors.

Use case	Approach
Nearby providers	ST_DWithin with a GiST index.
Distance ranking	ST_Distance after candidate filtering.
Live tracking	Append provider_location_updates during active travel.
Pickup location	Persist selected coordinates in booking_locations.
Privacy	Retain detailed GPS history only as long as justified.

19. Data Integrity & Transaction Rules

Rule	Implementation
One active assignment	Transaction/row lock or equivalent concurrency control for provider assignment.
Vehicle ownership	Ensure customer_id owns vehicle_id before booking creation.
Estimate approval	Only the booking customer can approve.
Invoice consistency	Invoice totals reconcile with invoice_items and payments.
Review eligibility	Only eligible completed bookings can be reviewed.
Status history	Every status mutation writes history in the same transaction.
Provider verification	Dispatch excludes non-VERIFIED/ACTIVE providers.
Financial deletion	Never hard-delete financial records; use void/reversal states.

20. Security & Privacy Controls

Area	Control
Authentication	Strong password hashes and secure session/OTP mechanisms.

Area	Control
Authorization	RBAC plus object-level access checks.
Documents	Private object storage with short-lived signed URLs.
Payments	Store gateway references, not raw card credentials.
Location	Limit live location access to relevant active jobs.
Audit	Record sensitive administrative/financial changes.
Uploads	Validate type/size and scan uploaded files.
PII	Minimize collection, restrict access, encrypt, and define retention.

21. Recommended Database Architecture

For the initial product, use one PostgreSQL database with clear modular domain ownership. A modular monolith is simpler to operate while preserving a clean path to later service separation.

Module	Tables
Identity	users, user_addresses
Vehicle	vehicles
Provider	providers, provider_documents, provider_services, provider_availability, provider_location_updates
Catalog	services
Booking/Dispatch	bookings, booking_locations, booking_status_history
Job	job_cards, inspections, inspection_items
Commercial	estimates, estimate_items, invoices, invoice_items, payments, payouts
Engagement	reviews, notifications, messages
Support/Governance	support_tickets, audit_logs

22. Backup, Recovery & Retention

Requirement	Recommendation
Backups	Automated daily backups plus point-in-time recovery where available.
Recovery	Regular restore testing.
Financial records	Retain according to applicable accounting/tax/legal requirements.
Location history	Shorter retention unless operational/legal reasons require longer.
Audit logs	Longer retention where needed for security and disputes.
Documents	Lifecycle policies and secure deletion at retention end.

23. Future Extensions

Capability	Likely additions
AI diagnosis	diagnostic_sessions, diagnostic_findings, recommendations.
Parts marketplace	parts, suppliers, inventories, purchase_orders, part_order_items.
Garage management	garage_staff, work_bays, inventory_items, stock_movements.
Fleet management	fleets, fleet_members, fleet_vehicles, fleet_service_policies.
Insurance	insurance_policies, claims, claim_events.
EV ecosystem	charging_stations, charging_sessions, battery_health_records.
Subscriptions	plans, subscriptions, subscription_benefits, usage_records.

24. Final Schema Summary

RoadResQ's database centers on a strong transactional chain: **users** → **vehicles** → **bookings** → **providers** → **jobs** → **estimates** → **invoices** → **payments**. Historical tables preserve state changes and operational evidence, while PostGIS supports location-aware provider matching.

The schema is suitable as an MVP database blueprint and is extensible toward garage management, fleet services, parts, insurance, EV support, subscriptions, and AI-driven automotive intelligence.